# DESIGN PROJECT CHALLENGES

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# Important note

Within this course, we want to work towards proofs of concept that tackle a real world or wicked challenge with a clear societal or economic impact. Many of these challenges contribute to one or more of the sustainable development goals (SDGs), while others start from a concrete need of a company, organisation or research group. We offer you several challenges which focus on different domains and aspects. These challenges are the starting point of the first phase of the first semester, namely the Design Thinking phase. While challenges push you in a certain direction, please read them with an open mind. In this course, we want you to go further than the initial assignment and look at ideas from many different angles. Let the challenges inspire you and come up with ideas that can brighten our society and tackle real world problems.

For most challenges, you can consider different possible perspectives/viewpoints and how this affects a possible technological solution:

- How can we solve this challenge for adults, children, students, companies, and health care professionals?
- Will our solution be proactive (avoid the challenge) or mitigate the consequences of the challenge?
- Think big! During the first steps of the Design Thinking phase, try to think outside the box. It can help to assume a world with no technological limits. Afterwards, in the following steps & phases, you can focus on scaling down towards a feasible product.
- Think about the technical difficulty of the proposed solution. As this is still a course in the Master of Science in Computer Science Engineering, aspects of technical depth and difficulty need to be present to develop the competencies linked to this course.

# 1. Resilient AI: Autonomous AI Data Centers

**Context**

This work will be supported by InnoEnergy, early-stage impact investor in climate tech.

AI has become a driving force of the energy transition, our industries and the new economy. Its growing power needs demand cleaner, cheaper energy, while its potential to accelerate innovation is unmatched. From precision weather and demand-response forecasting to autonomous optimisation, AI will transform how we produce, distribute, and consume energy. This is exactly where InnoEnergy is focusing its AI efforts: apply AI to enable the energy transition and make AI sustainable, available and resilient.

It becomes clear that the AI data centers are becoming a critical resource as they run a range of mission critical services from supporting emergency services, executing financial transactions, support hospitals, run industrial control and distributed energy systems, air traffic control and many many more. AI data centers need guaranteed power to operate: this is not only about the availability of energy to run the AI data center but also the resilience of AI data centers following the loss or the degradation of its power and energy supply.

**Challenge**

The mission-critical nature of AI is where energy resilience becomes interesting. Suppose the power grid connection suddenly disappears. The future resilient AI data architecture should be one where AI predicts the available energy, the workloads are reprioritised, cooling is adjusted, batteries dispatched and non-critical workloads migrated.

A software management or operations platform should calculate the optimal time remaining to deliver the AI service or decide which tasks that need to migrate in order to keep critical services ON as long as possible. This approach changes traditional data center resilience from hardware redundancy into dynamic operational resilience.

We want you to design and implement an autonomous software platform that enables a critical AI data centre to maintain maximum mission-critical compute capacity for 24–72 hours following loss or severe degradation of its external power supply, by dynamically orchestrating workloads, batteries, generators, cooling and grid interaction.

**Contact**

prof. dr. ir. Frank Gielen (frank.gielen@innoenergy.com)

# 2. Designing Green Within the Rules: Real-Time Sustainability Feedback for Medical Devices

**Context**

The medical sector and more specifically medical devices, a broad group of all products regulated under the Medical Device Regulation (MDR), accounts for a significant share of global greenhouse gas emissions, chemical pollution, and waste. Despite a growing willingness within the medical sector to adopt more environmentally friendly solutions, compliance with regulatory requirements or guidelines that prioritize patient safety often complicates initiatives aligned with EU sustainability policies and frameworks. As a result, conflicting demands may arise for medical devices (e.g., integrating circular principles such as reusing/recyclable products can conflict with strict legislative hygiene and infection prevention requirements).

Life Cycle Assessment (LCA) software can enable quantifying the impacts of these products during the development process. However, today where it is used within product development, it tends to be deployed either at the very beginning of a project (to benchmark competing products or test a concept) or at the very end (as a final evaluation of the finished design's environmental impact). It is also not adapted to the medical device design context: the development stages and information available at each phase, the medical and regulatory constraints, and the assumptions that should reflect actual healthcare practice. In both cases, LCA functions as a snapshot rather than a continuous instrument, and plays little or no role in the many individual design choices made throughout development. Current LCA tools do not facilitate the moments when a design's environmental footprint is actually determined: the countless material, component and process decisions made throughout development. By the time a benchmark or end-of-project evaluation is run, those choices are already locked in. This project addresses that gap by making sustainability evaluation a constant part of the medical device development process itself. The project will be executed and evaluated throughout the D4SMD project (https://sustamed.ugent.be/Projects/D4SMD.html)

**Challenge**

How might we enable medical device development teams to continuously evaluate the sustainability of their design choices throughout the entire development process?

**Target audience:** Medical device design teams

**Contact**

Pieter Beerten (Pieter.beerten@ugent.be)

# 3. The Future of Fan Experience

**Context**

Our digital lives have taken unprecedented jumps over the past decades, but somehow, our customer experience in physical venue visits has not fully adopted the digital opportunities yet. We still struggle to find the right entrance to our seat in a stadium, we are all still at the slowest entrance (queue), we are all at the busiest toilets while our friends took only half the time we did. Event venues (football stadiums, concert halls, festivals) tend to see Wi-Fi as an obliged cost, but miss to see the tremendous opportunity it can hold, if used in an intelligent way. The Future of Fan experience is hybrid – a physical visit enriched with a digital layer that does not distract from the event, but nicely complements to it and enriches the experience.

**Challenge**

We challenge you to design a digital tool/platform that allows the owners of a venue (Let's take a Sports stadium as an example) to analyze & understand their venue better and communicate directly to the end users with personalized and contextualized messages that truly bring value to the experience of the visitor – so to create the **Fan Visit Experience 2.0.** You can assume that all visitors will be connected to the Stadium's WiFi network throughout their full stay/visit

Things to consider:

- Think how a digital management platform for Stadium owners can be used to manage, enrich or (re)design the digital aspect of the physical visitor journey from arrival, to event, to departure...
- Assume all visitors are connected to the Stadium's Enterprise Wi-Fi throughout their visit
- To properly enrich and bring value, the Stadium host needs to be able to communicate on an individual, contextual basis with the visitors, so the visitor always perceives it positively
- No visitor is the same – think about how the stadium owner can identify patterns and use this to classify visitors by their preferences – to customize the offer to them even better.
- The platform should provide advanced intelligence and communication features while remaining intuitive and agile for non-technical personnel (commercial managers, operations managers,…)
- Respect privacy-by-design principles and comply with GDPR.
- Connectivity is not flawless and gets interrupted – ensure that the solution can deal with brief interruptions or connectivity loss, without impacting the visitor journey
- To avoid a user being bombarded with conditional messaging campaigns, think about how the total frequency of interaction can be optimized and controlled by the user of your solution

**Target audience**

The digital solution should be used by the commercial (marketing & sales) and operations management of the venue owner. The visitors will only receive the produced outcome of this platform as communications / pushes on their mobile device to enrich their experience

**Contact**

Nicolas Van Oevelen (Nicolas.vanoevelen@roow.io) & Louis Muys (Louis.Muys@roow.io)

# 4. The Patient's Journey of the Future

**Context**

Healthcare has become highly digital, yet the patient experience did not. Patients wait without information, visitors struggle to navigate, and caregivers spend valuable time answering routine questions instead of providing care. What if every patient and visitor could enjoy a seamless, personalized digital journey, nurses can communicate with patients seamlessly, while hospital staff gain live operational insights to optimize resources, patient flow and quality of care.

**Challenge**

Design a digital platform that enables the hospital staff/management to orchestrate and transform the hospital experience for patients and visitors. Think how we can create a digital platform that combines live operational insights, intelligent workflows and personalized communication to improve patient care, visitor experience and hospital efficiency - assuming patients and visitors already are all automatically, seamlessly connected on the hospital's enterprise Wi-Fi network from the moment of their arrival till the moment they leave.

Things to consider:

- Think how a digital management platform for Hospital staff/managers can be used to (re)design/optimize the complete patient journey from arrival and digital check-in through admission, treatment, recovery and discharge,
- The platform should provide advanced operational intelligence and communication features while remaining intuitive and agile for non-technical personnel (doctors, nurses, admin staff, …)
- Real-time updates on appointments, meals, doctors, treatments and discharge.
- The platform should support two-way communication between patient and caregiver, based on location, appointment status, patient profile, department, live operational events, …
- Think about how features can also help the journey for visitors (non-patient)
- Respect privacy-by-design principles and comply with GDPR and healthcare regulations while protecting sensitive medical information.
- You can assume all patients and visitors are connected to the Hospital WiFi throughout their stay
- Which relevant data can be pulled out of this connected journey that can help staff, hospital management, … to make better decisions on operations, logistics, staffing, …

Create a solution for Hospitals that allows them to manage and control the patient's customer journey to make a hospital stay/visit digital and intuitive. Think big, challenge existing patient

journeys and design a digital tool that enables non-tech users to create the Patient Journey 2.0 – improving the lives of patients, visitors, caregivers and hospital management!

**Contact**

Nicolas Van Oevelen (Nicolas.vanoevelen@roow.io) & Louis Muys (Louis.Muys@roow.io)

# 5. What just happened out there? Helping a small sports club learn from its own game

**Context**

In a club that runs mostly on volunteers, almost everything known about the different teams of each league (separate girls and boys leagues, grouped per 2 years), lives in someone's head. A coach sees a match once, from the side of the pitch, and remembers a handful of moments. Players hear "you were standing too deep" and have no way to check what that actually looked like. Youth coaches change from season to season and the picture of how a player has developed starts from zero each time. Meanwhile matches are filmed, statistics are being written on paper, and almost none of it comes back to the people it is about.

Belgian hockey has grown to roughly 60,000 members (of which 70 – 80% youth players) across more than 100 clubs, the majority of those clubs are small but with some budget and sponsors, the larger clubs with a bigger budget and sponsors. To give an idea, typical Belgian youth hockey membership fees are often in the range of approximately 400 - 500 EUR, which adds up to a substantial annual membership market, depending on the club, facilities, and age category. (This is an estimate based on public club fee structures, not a federation statistic.)

**What already exists:** Professional platforms (Nacsport, Pixellot, Spiideo, Catapult and others) do this for top sport, with hardware, budget and an analyst. The club also built a small proof of concept on its own footage: automatic player tracking works reasonably well, ball detection is much harder. That report, sample footage and the technical explorations are available to you, as material to build on or to walk away from. Video and AI are one possible route, not the assignment.

**Challenge**

How can a small club, with mostly volunteers, one camera and a small budget, actually learn from what happens on its pitch, and give coaches and young players something they can act on? That question is deliberately left open. Who you help is part of the problem: the head coach, the individual player, the youth academy, a parent, the board. What "learning from the game" means is part of it too, from tactical patterns to individual progress over a season, motivation, playing time, or spotting a player who is quietly getting better.

**The one hard constraint:** Whatever you design has to be usable by a club that has no analyst, a limited budget and no specialised hardware. If it only works for the top, it does not solve this problem.

**What the challenger brings:** Meetjesland Hockey Club (Oost-Vlaanderen) brings a real club, real match footage, and coaches and players as end users. Its sports director Juan Montone leads the T1 men and is an international player and coach, so he will tell you honestly whether what you built is useful. The organisation is small enough to decide quickly and test with you during the season. Additionally, as hockey is still considered an elite sport, the bigger clubs have budgets. And although they are not using the professional analytical tools (we know this from Juan Montone, who also coaches in a larger club), it is likely that they would be interested in such a tool for their club. Such a tool has international potential.

**Contact**

Dr. Inge de Waard, AI & Learning Lead at InnoEnergy, club member and hockey parent (inge.dewaard@innoenergy.com)

# 6. Why Should This Work? Building Digital Health Interventions That Carry Their Own Theory

**Context**

Thousands of health apps promise to help us move more, sleep better, drink less or quit smoking. Most are abandoned within weeks, and very few can explain why they should work. Features such as streaks, badges, reminders and step counters are often added because they are easy to build or popular. They are rarely chosen because they target what actually keeps a person from changing.

Behavioural science offers another route. First you analyse what drives a behaviour: capability, motivation, opportunity, habits, and the social and physical context. Then you select behaviour change techniques (BCTs) that target those drivers. Finally, you make the expected chain of effects explicit in a program theory.

In practice, this reasoning lives in reports and slide decks, disconnected from the software. Once the app is built, nobody can trace which feature was supposed to change which determinant, or whether it did. Nobody can see whether it also produced unintended effects such as guilt, social pressure, compensation or disengagement (so-called dark logic).

**Challenge**

How might we build a digital health intervention in which the program theory is not a document but part of the system? Every feature should be traceable to the determinant it targets, and the system should learn whether that assumption actually holds.

Start from a concrete behaviour in a concrete group. Examples are sleep, active commuting, alcohol use, or taking study breaks. Some key questions to consider:

- How can you represent determinants, techniques and features in a machine-readable way, so that design choices become explicit and queryable? You could build a knowledge graph on existing work such as the Behaviour Change Intervention Ontology.
- How can the system tailor which technique is delivered, when, and to whom, based on context and the person's own data (just-in-time adaptive interventions)?
- How can you build in experimentation, so the system tests its own program theory instead of assuming it? Options include micro-randomised decisions, short in-the-moment questionnaires (EMA) and logging of what users were actually exposed to.
- How can you detect and monitor unintended effects, and not only the intended ones?
- How can you present this evidence to intervention designers so it helps them decide what to keep, adapt or drop?

- How do you deal with the privacy and ethics of continuous behavioural and contextual data?

We are not looking for yet another app with badges. The value lies in the link between the why and the what.

**Target audience and intervention**

**Target population:** Adults who want to change a specific health behaviour, but think beyond the straightforward groups such as students. A second group is health promotion professionals who design and evaluate interventions.

**Target for intervention:** One health behaviour of your choice. Also the design process itself: making the reasoning behind digital interventions explicit, testable and revisable.

**Setting:** Daily life, via smartphone, wearables and the campus environment, combined with a designer-facing dashboard or tool.

**Contact**

Geert Crombez (geert.crombez@ugent.be)

# 7. Nudges That Don't Nag: Breaking Up Prolonged Sitting Without Breaking the Flow

**Context**

Office workers often sit for hours without a break, although short interruptions (standing up every 30 minutes or so) bring clear health benefits. Reminders and prompts seem an obvious solution. However, health promotion students who designed such an intervention found a problem through their dark logic analysis: fixed reminders are experienced as disruptive or patronising. They break concentration and clash with work pressure. People start ignoring them, and the cue loses its effect. In addition, success was only tracked by how often messages were opened, not by whether people actually stood up.

**Challenge**

How might we design a prompting system that knows when a person can be interrupted, adapts to the person, and gradually makes itself unnecessary as the new habit forms?

Some questions to consider:

- How can you detect prolonged sitting and interruptible moments with low burden? Possible sources are sensors, keyboard activity, calendar or wearable data.
- How can the system learn which timing and type of prompt works for whom, without becoming annoying?
- How can cues be faded out over time, so that behaviour becomes habitual rather than prompt-dependent?
- How do you measure actual behaviour, and unintended effects such as irritation or reduced concentration, rather than message opens?
- How can social norms at the workplace be made visible (e.g., colleagues or managers standing up) without creating pressure?

**Target audience and intervention**

**Target population:** Adults with predominantly sedentary jobs (office, IT, administration).
**Setting:** Workplace, including hybrid and home office.

**Contact**

Geert Crombez (geert.crombez@ugent.be)

# 8. Supporting Physical Activity in People with Epilepsy

**Context**

Regular physical activity has important physical and mental health benefits for people with epilepsy. Nevertheless, people with epilepsy are generally less physically active than the general population, with approximately 65% of adults with active epilepsy not meeting recommended levels of moderate-to-vigorous physical activity. People with epilepsy may experience specific barriers to being active, including fear of having a seizure during physical activity, concerns about injury and safety, stigma or embarrassment, limited social support, and overprotective attitudes from others. These barriers can also depend strongly on the situation. Crowded, noisy or unfamiliar environments may increase stress and seizure-related anxiety, whereas familiar, calm environments and the presence of trusted people can increase feelings of safety and confidence.

Technology could offer new ways of providing personalized and context-sensitive support in daily life, for example by using smartphones, wearables or contextual information to adapt support to a person's needs and circumstances.

**Challenge**

How might we design a technological solution that supports people with epilepsy in becoming more physically active in daily life, while taking into account seizure-related concerns, perceived safety, social support and changing daily-life contexts?

The system could combine information from smartphones, wearables or user input, such as physical activity, recent seizure occurrence, location, stress, fatigue or perceived safety. Based on this information, it could adapt the timing and content of its support, for example by suggesting a suitable activity, a safe or familiar environment, social support, or by recognizing moments when physical activity encouragement is not appropriate.

Some questions to consider:

- How can technology provide personalized support at the right moment, rather than using a one-size-fits-all approach?
- How can data from smartphones, wearables, sensors and user input be combined to determine when and how support should be provided?
- How can the system support feelings of safety and confidence without unnecessarily restricting physical activity?
- How can privacy, user burden and individual differences be taken into account?

**Target audience and intervention**

**Target population:** Adults with epilepsy

**Target for intervention:** Supporting safe and sustainable physical activity in everyday life through personalized and context-sensitive digital support.

**Contact**

Julie Delobelle (Julie.Delobelle@UGent.be) & Delfien Van Dyck (Delfien.VanDyck@UGent.be)

# 9. How to support muscle-strengthening activities in older adults?

**Context**

Regular physical activity contributes to healthy ageing by improving physical function, maintaining independence, and reducing the risk of chronic diseases and falls. According to the World Health Organization (WHO), older adults should engage in at least 150 to 300 minutes of moderate-intensity aerobic physical activity per week. This guideline is relatively well-known by older adults. However, the WHO also recommends older adults to perform muscle-strengthening activities on at least two days per week. Only a very small proportion of older adults (approximately 10%) in Flanders currently meet the guidelines for muscle-strengthening activities.

The research unit of prof. Van Dyck recently conducted a large-scale survey study in older adults (65+) on this topic, collecting information of almost 2000 older adults across Belgium. This study showed that older adults generally recognize the benefits of strength exercises and are motivated to maintain their health and independence. However, many are uncertain about what muscle-strengthening exercises actually involve, how to get started, where such activities can be performed, and whether special equipment is required. In addition, older adults prefer to exercise together with others, value age-appropriate activities, and may face financial barriers when participating in organized programs.

Technology could offer new ways to support older adults in discovering, initiating, and maintaining muscle-strengthening activities.

**Challenge**

How could a technological solution support older adults in initiating and maintaining muscle-strengthening activities?

Some questions to consider:

- How can technology help older adults understand what muscle-strengthening activities are and how they can get started?
- How can information from user preferences, location, physical abilities, and available community resources be used?
- How can social support and participation be supported?

**Target audience and intervention**

**Target population:** Adults aged 65 years and older

**Target for intervention:** Supporting the adoption and long-term maintenance of muscle-strengthening activities among older adults

**Contact**

Delfien Van Dyck (Delfien.VanDyck@UGent.be)

# 10. How can we keep children going with their balance exercises at home?

**Context**

The vestibular organs, located in the inner ear, play an important role in keeping us upright, stabilising our gaze when we move our head and helping us understand where we are in space. When that system does not work well in a child, the consequences reach far beyond dizziness: delayed motor development, poor balance, trouble concentrating, and children who slowly drop out of play and sport. Based on current published prevalence estimates, between approximately 400 and 5600 children aged 4 to 18 years in Flanders may have a vestibular dysfunction.

The standard treatment is vestibular rehabilitation therapy, an exercise-based therapy delivered by physiotherapists. Depending on the child's specific vestibular problem, developmental level and functional goals, therapists prescribe individually adapted balance and gaze-stability exercises involving head, eye and body movements. These exercises are practised both during supervised therapy sessions and at home, often regularly over an extended period. Consistent home practice is an essential part of the rehabilitation process. However, maintaining engagement and adherence at home is challenging, particularly for young children. Exercises can be repetitive, symptom-inducing or difficult to integrate into daily routines. Currently, home exercises are largely supported through paper-based instructions, which provide limited support for motivation, understanding how to perform the exercises and why they are important, or for parents in supporting their child. Therapists also have limited insight into how consistently exercises are performed, how children experience them or whether difficulties arise between appointments.

**Challenge**

How might we use technology to help children stay engaged with their prescribed vestibular rehabilitation exercises at home, while helping parents understand and support the exercises and giving therapists better insight into what happens between appointments?

**Target audience and intervention**

**Target population:** Children roughly 4 to 18 years old with a vestibular dysfunction, plus their parents, who supervise the exercises at home, and their therapists. Throughout this course, the target population can be narrowed based on the insights that are gained.

**Target for intervention:** Motivation, engagement and adherence to the home exercise program, correct execution of the exercises, and the insight a therapist can build up remotely.

**Contact**

Femke De Backere (femke.debackere@ugent.be) & Ruth Van Hecke (ruth.vanhecke@ugent.be)

This challenge comes from the BalanceCare4Kids research group of Ghent University and Ghent University Hospital, so domain experts and potential end users are within reach.

# 11. Asking the Right Questions: From Standard Questionnaires to Personally Relevant Measurement

**Context**

Health care and health research rely heavily on standardised questionnaires. Everyone answers the same fixed items, such as "I feel tense" or "Pain interferes with my daily activities", on the same rating scale. That makes it easy to compare people and pool data across studies. But it says little about what matters to this person, in their daily life. One person's stress comes from unexpected phone calls, another's from deadlines. One person's pain gets worse after sitting still for a long time in the morning, another's in the evening after a busy day. Generic items miss these personal triggers, situations and meanings, which are exactly what an intervention should target. Many items also feel irrelevant to respondents, so they disengage and give careless answers, especially when they are questioned repeatedly over days or weeks.

An alternative is idiographic measurement: items built from the person's own experiences and words. Examples are "My pain increases when I have to sit still for a long time in the morning" or "I feel tension when I get an unexpected phone call". Today these items are made by hand, in lengthy interviews with a clinician or researcher. This makes them expensive, hard to scale, and dependent on the interviewer. Conversational surveys could change this. They ask one question at a time in natural language and adapt their follow-up questions to previous answers, increasingly with the help of AI. They combine the depth of an interview with the scalability of a survey.

**Challenge**

How might we design a system that holds a natural conversation with a person, discovers what really matters in their situation, turns that into a small set of personal items in their own words, and then uses those items to follow that person over time?

Start from one concrete domain, such as pain, stress, sleep, fatigue or a health behaviour you want to change. Some key questions to consider:

- **Conversation:** How can a conversational agent ask open, non-leading questions, follow up adaptively on the answers ("You mentioned deadlines. What exactly about deadlines makes it difficult?"), and know when it has enough information?
- **From talk to items:** How can situations, triggers, emotions and behaviours be extracted from the conversation and turned into clear, measurable items that keep the person's own language?

- **Person as co-author:** How can people check and adjust their items ("Does this sentence fit how you experience it? Would you phrase it differently?") so they recognise their own experience in them?
- **Linking to theory:** How can personal items stay connected to underlying concepts, such as confidence, avoidance, triggers or habits, so that measurements remain interpretable and to some extent comparable between people? A structured vocabulary or ontology could help here.
- **Repeated measurement (N=1):** How can personal items be delivered repeatedly in daily life, at the right moments and with low burden, and turned into feedback that a person or clinician can act on?
- **Quality and risks:** How do you know whether an item is reliable and actually measures what it should? How do you avoid an AI agent that steers answers, invents themes or misreads what someone means? How do you protect sensitive personal information?

We are not looking for a chatbot that simply administers an existing questionnaire. The value lies in making measurement personally relevant without losing rigour.

**Target audience and intervention**

**Target population:** People who want to monitor or change a health-related experience or behaviour, such as people with persistent pain, stress or sleep problems, or students as an accessible starting group. A second group is clinicians and researchers who want personalised assessment.

**Target for intervention:** The measurement process itself: how questions are generated, validated and used to follow individuals over time and to guide personalised interventions.

**Setting:** Daily life, via smartphone or web, combined with a clinician- or researcher-facing overview.

**Contact**

Annick De Paepe (annick.depaepe@ugent.be)

# 12. Accurate dietary assessment

**Context**

Diet plays a pivotal role in individual and population health, with healthy dietary patterns increasingly recognised as an important determinant of healthy ageing and the prevention of chronic diseases. Accurate dietary assessment is therefore essential to gain insight into eating behaviours, to evaluate nutrient adequacy, and to inform the development of personalized nutrition interventions.

Dietary intake is commonly assessed using self-reported methods, such as 24-hour recalls (24HRs), food frequency questionnaires (FFQs), and food records. While these methods can provide valuable information, they are often associated with substantial participant burden and are prone to reporting errors. Individuals may forget foods, inaccurately estimate portion sizes, or have difficulty identifying the appropriate food item in extensive food composition databases.

Recent developments in mobile technologies, image-based dietary assessment, artificial intelligence, and wearable sensors offer new opportunities to support dietary recording. However, important challenges remain, particularly for **composite dishes**, **local specific foods**, and the need to balance accuracy with ease of use. Technological solutions that minimize user burden while maintaining high-quality dietary data could substantially improve both research and practice.

**Challenge**

How might we design a technological solution that enables accurate and user-friendly dietary assessment (tailored to the Flemish context) while minimizing participant burden?

Some questions to consider:

- How can technology reduce the burden of dietary recording without compromising data quality?
- How can local dishes, homemade meals, and culturally specific foods be accurately represented within dietary assessment systems?
- How can composite dishes accurately be recorded, while minimising user burden?
- How can dietary assessment systems be adapted to different populations, such as older adults, or individuals with low digital literacy?

**Target audience and intervention**

**Target population:** Adults in the general population, with a particular focus on middle-aged and older adults.

**Target for intervention:** Supporting accurate dietary assessment in everyday life while minimising user burden and facilitating the recording of complex meals and local food items.

**Contact**

Kim Cnudde (Kim.Cnudde@UGent.be) & Delfien Van Dyck (Delfien.VanDyck@UGent.be)

# 13. Detecting and Understanding Autism-Related Movement Patterns

**Context**

Early detection of autism characteristics remains a major challenge. Some of the earliest observable indicators of autism appear in children's everyday behaviour, such as restrictive and repetitive behaviours (RRBs), atypical responses to sensory input, and differences in play, interaction and communication. Technology can help detect such behaviours objectively and early, for example through wearable sensors or automated video analysis.

A crucial step in developing and validating such detection systems is high quality reference data. To develop and evaluate machine learning algorithms, researchers first need accurately annotated video recordings that indicate exactly when specific behaviours occur. Today this process is time consuming: human observers have to code children's behaviour from video recordings systematically. Once behaviours are coded, researchers can use methods to analyse the resulting data, explore patterns, quantify behavioural characteristics and derive insights that can be used to improve detection algorithms.

**Challenge**

This challenge addresses two key research and engineering problems: (1) developing a robust system for coding behaviour from video data and (2) developing methods to analyse and visualise the coded behavioural data.

The goal is to go beyond simple video annotation and create an integrated workflow that supports efficient behavioural coding, quality control and advanced data analysis.

Students will work with video recordings of young children and with coding schemes that describe autism related behaviours. The system should be flexible enough to support different coding schemes and types of behaviour.

Some questions to consider:

- How can you make the annotation process faster and less burdensome for researchers?
- How can timestamps and behavioural events be captured efficiently?
- How can the system support different coding schemes and adapt when a scheme changes?
- How can partial automation or coding assisted by AI support human annotators?
- How can the system facilitate quality control and checks on reliability between coders?
- Which behavioural metrics are most informative for characterising a child's behaviour?
- How can coded data be turned into meaningful visualisations and dashboards?

- How can the coded data be combined with other data sources, such as sensor data, and prepared for future machine learning models that detect behaviours automatically?

**Target audience**

Researchers in a variety of contexts that need to code behaviours in recording

**Contact**

Sofie Boterberg (sofie.boterberg@ugent.be)